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Cryptology Assignment #3

1. Determine the multiplicative inverse modulo 26 for each of the following integers.

i) 1 → $26 = 1(26) + 0$

$1(1) = 1$

$1^{-1} = 1 \pmod{26}$

Multiplicative inverse of 1 = 1

ii) 3 → $26 = 3(8) + 2$

$3 = 2(1) + 1$

$1 = 3 - 2$

$2 = 26 - 3(8)$

$1 = 3 - (26 - 3(8))$

$1 = 3 - 26 + 3(8)$

$1 = 3(9) + 26(-1)$

$3^{-1} = 9 \pmod{26}$

Multiplicative inverse of 3 = 9

iii) 5 → $26 = 5(5) + 1$

$1 = 26 - 5(5)$

$1 = 5(5) + 26(1)$

$-5 + 26 = 21$

$5^{1} = 21 \pmod{26}$

Multiplicative inverse of 5 = 21

iv) $7 \rightarrow 26 = 7(3) + 5$, $5 = 26 - 7(3)$

$7 = 5(1) + 2$, $2 = 7 - 5$

$5 = 2(2) + 1$

$1 = 5 - 2(2)$

$2 = 5 - 2(7-5)$

$1 = 5 - 2 \times 7 + 2 \times 5$

$1 = 3(5) - 2(7)$

$1 = 3(26 - 7(3)) - 2(7)$

$1 = 3 \times 26 - 3 \times (7(3)) - 2(7)$

$1 = 3 \times 26 - 11 \times 7$

$1 = (-11)7 - 3 \times 26$, $-11 + 26 = 15$

$7^{-1} = 15 \pmod{26}$

Multiplicative inverse of $7 = 15$

v) $9 \rightarrow 26 = 9(2) + 8$

$9 = 8(1) + 1$

$1 = 9 - 8$, $8 = 26 - 9(2)$

$1 = 9 - (26 - 9(2))$

$1 = 9 + 9(2) - 26$

$1 = 3(9) - 26$

$9^{-1} = 3 \pmod{26}$

Multiplicative inverse of $9 = 3$

vi) $11 \rightarrow 26 = 11(2) + 4$

$11 = 4(2) + 3$, $3 = 11 - 4(2)$

$4 = 3(1) + 1$, $4 = 26 - 11(2)$

$1 = 4 - 3$

$1 = 4 - (11 - 4(2)) = 4 - 11 + 4(2)$

$1 = 3(4) - 11$

$1 = 3(26 - 11(2)) - 11$

$1 = 3 \times 26 - 3 \times (11(2)) - 11$

$1 = 3 \times 26 - 7 \times 11$

$1 = (-7)11 + 3 \times 26$, $-7 + 26 = 19$

$11^{-1} = 19 \pmod{26}$

Multiplicative inverse of 11 is 19

vii) 15 → $26 = 15(1) + 11$, $11 = 26 - 15$

$15 = 11(1) + 4$, $4 = 15 - 11(1)$

$4 = 3(1) + 1$, $3 = 11 - 4(2)$

$1 = 4 - 3$

$1 = 4 - (11 - 4(2))$

$1 = 3(4) - 11$

$1 = 3(15 - 11(1)) - 11$

$1 = 3 \times 15 - 4 \times 11$

$1 = 3 \times 15 - 4(26 - 15)$

$1 = 7(15) - 4(26)$

$15^{-1} = 7 \pmod{26}$

Multiplicative inverse of $15 = 7$

viii) 17 → $26 = 17(1) + 9$, $9 = 26 - 17(1)$

$17 = 9(1) + 8$, $8 = 17 - 9(1)$

$9 = 8(1) + 1$

$1 = 9 - 8$

$1 = 9 - (17 - 9(1))$

$1 = 2(9) - 17$

$1 = 2(26 - 17(1)) - 17$

$1 = 2 \times 26 - 2 \times (17(11) - 17)$

$1 = 2 \times 26 - 13 \times (17)$

$1 = (-3)17 - 2(26)$, ~~$17^{-1} = 23$~~

$17^{-1} = 23 \pmod{26}$

Multiplicative inverse of 17 = 23

ix) 19 → $26 = 19(1) + 7$, $7 = 26 - 19$

$19 = 7(2) + 5$, $5 = 19 - 7(2)$

$7 = 5(1) + 2$, $2 = 7 - 5(1)$

$5 = 2(2) + 1$

$1 = 5 - 2(2)$

$1 = 5 - 2(7 - 5(1))$

$1 = 5 - 2 \times 7 + 2(5(1))$

$1 = 3(5) - 2(7)$

$1 = 3(19 - 7(2)) - 2(7)$

$1 = 3 \times 19 - 8(7)$

$1 = 3(19) - 8(26 - 19)$

$1 = 11(19) - 8(26)$

$19^{-1} = 21 \pmod{26}$

Multiplicative inverse of 19 = 11

x) 21 → $26 = 21(1) + 5$, $5 = 26 - 21$

$21 = 5(4) + 1$

$1 = 21 - 5(4)$

$1 = 21 - 4(26 - 21)$

$1 = 21 - 4(26) + 4(21)$

$1 = 5(21) - 4(26)$

$21^{-1} = 5 \bmod 26$

Multiplicative inverse of 21 = 5

xi) 23 → $26 = 23(1) + 3$, $3 = 26 - 23$

$23 = 3(7) + 2$, $2 = 23 - 3(7)$

$3 = 2(1) + 1$

$1 = 3 - 2(1)$

$1 = 3 - (23 - 3(7))$

$1 = 8(3) - 23$

$1 = 8(26 - 23) - 23$

$1 = (-9)23 + 26(3)$

$26 - 9 = 17$

$23^{-1} = 17 \bmod 26$

Multiplicative inverse of 23 = 17

xii) 25 → 26 = 25(1) + 1

1 = 26 - 25

1 = (-1)25 + 26

gl - l = 25

$25^{1} = 25 \bmod 26$

multiplicative inverse of 25 = 25

2. choose any string of your choice to encrypt using the affine cipher with key values multiplier 5 and additive 8.

$$C = E(K_1 * P + K_2) \bmod 26$$

$K_1 = 5$, $K_2 = 8$

PT		$K_1 * P$	$+ K_2$	mod 26	
G	6	30	38	12	M
O	14	70	78	0	A
O	14	70	78	0	A
D	3	15	23	23	X
L	11	55	63	11	L
U	20	100	108	4	E
C	2	10	18	18	S
K	10	50	58	6	G

Cipher text = MAAXLESG

3. Encrypt "HI" using the Affine cipher with a = 11 and b = 5

$$C = E(K_1 * P + K_2) \bmod 26$$

$K_1 = 11$, $K_2 = 5$

PT		$K_1 * P$	$+ K_2$	mod 26	
H	7	77	82	4	E
I	8	88	93	15	P

Cipher text = EP

Cipher text VCR was encrypted using affine function ~~9x+2~~ $9x+2 \bmod 26$. Find plain text.

$$P = K_1^{-1}(C - K_2) \bmod 26$$

$K_1 = 9$, $K_2 = 2$

$9^{-1} \bmod 26 = 3$

~~$26 = 9(2) + 8$, $8 = 26 - 9(2)$~~

CT		$K^{-1}(C-K_2)$	mod 26	PT
V	20	$3(20-2) = 54$	2	C
C	2	$3(2-2) = 0$	0	A
R	17	$3(17-2) = 45$	19	T

Plain text = CAT

Decrypt the cipher text using Affine cipher, KDYXJVWDEXZY using key 3 (mul) & 4.

$$P = K_1^{-1}(C - K_2) \bmod 26$$

CT = KDYXJVWDEXZY, $K_1 = 3$, $K_2 = 4$, $3^{-1} \bmod 26 = 9$

CT		$K_1^{-1}(C-K_2)$	mod 26	PT
K	10	54	2	C
D	3	-9	17	R
Y	24	181	24	Y
X	23	171	15	P
J	9	45	19	T
V	20	144	14	O
W	22	162	6	G
D	3	-9	17	R
E	4	0	0	A
X	23	171	15	P
Z	25	189	7	H
Y	24			Y

plaintext = CRYPTOGRAPHY

6. Use $K_1 = 13$ (mul), $K_2 = 4$ (Even though it is not a valid key but to encrypt).

Encrypt the string "input" & "alter" using above key & find the what is the issue.

$$C = E(K_1 * P + K_2) \bmod 26 \quad , K_1 = 13, K_2 = 4$$

PT		$K_1 * P$	$+ K_2$	~~mod 26~~	
I	8	104	108	4	E
N	13	169	173	12	R
P	15	195	199	17	R
U	20	260	264	4	E
T	19	247	251	17	R
A	0	0	4	4	E
L	11	143	147	17	R
T	19	247	251	17	R
E	4	52	56	4	E
R	17	221	225	17	R

$K_1 = 13$ is not co prime with 26

7. Find the Inverse Matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$

$$\begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$$

$11^{-1} \bmod 26 = 19$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8. Find the inverse of the following matrix $\begin{matrix} 11 & 13 \\ 2 & 3 \end{matrix}$

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = (11 \times 3) - (13 \times 3) = (33 - 26) = 7$$

$$7^{-1} \bmod 26 = 15$$

$$15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9. Is the key valid ? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} = (2 \times 21) - (13 \times 14) = 42 - 182 = 140$$

Invalid 140 is not coprime to mod 26

10. Encrypt using 2×3 Hill cipher for "SOLVED" using $\begin{matrix} 3 & 8 \\ 7 & 11 \end{matrix}$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = (3 \times 11) - (7 \times 8) = (33 - 56) = 23$$

23 is no zero

23 is co-prime to 26

PT = SOLVED

$$P = \begin{bmatrix} S \\ O \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} 3 \times 18 + 8 \times 14 \\ 7 \times 18 + 11 \times 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26 \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ U \end{bmatrix}$$

$$P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 12 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 12 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26 \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

$$P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \bmod 26 \begin{bmatrix} 10 \\ a \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

cipher text = KUTWKJ